10.1 Functions of two or more variables

Domain

polynomials: $f(x) = x^2$ $\quad Dom(f) = (-\infty, \infty)$

$= \mathbb{R}$

$f(x,y) = x^2 + y^2 + 3xy$ $\quad Dom(f) = (-\infty, \infty) \times (-\infty, \infty)$

$= \mathbb{R}^2$

$(-\infty, 0) \cup (0, \infty)$

$(-\infty, \infty) \times (-\infty, \infty) \times (-\infty, \infty) = \mathbb{R}^3$

logarithms: $f(x) = \ln(x)$ $\quad Dom(f) = (0, \infty)$

$f(x,y) = \ln(x^2 - y)$

$Dom(f) = \{(x,y) \in \mathbb{R}^2 \mid x^2 - y > 0\}$

$\{(x,y) \in \mathbb{R}^2 \mid y < x^2\}$

$\ln(x^2 + y^2 + 1)$

$\{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 + 1 > 0\}$

$\ln(x^2+y^2+1) = f(x,y)$

$\{(x,y) \in \mathbb{R}^2 \mid x^2+y^2+1 > 0\}$

$Dom(f) = \mathbb{R}^2 = (-\infty,\infty) \times (-\infty,\infty)$

Square roots: $\sqrt{x-3}$ $\quad x-3 \geq 0$

$\{x \in \mathbb{R} \mid x \geq 3\}$

$\sqrt{x^2+y-1} = f(x,y)$

$\{(x,y) \in \mathbb{R}^2 \mid x^2+y-1 \geq 0\}$

Rational functions: $f(x) = \frac{1}{x}$ $\{x \in \mathbb{R} \mid x \neq 0\}$

$(-\infty,0) \cup (0,\infty)$

$\frac{1}{x^2+3xy} = f(x,y)$

$Dom(f) = \{(x,y) \in \mathbb{R}^2 \mid x^2+3xy \neq 0\}$

Exponential: $\mathbb{R}$, $\mathbb{R}^2$

Trig: sin, cos, $\mathbb{R}$, $\mathbb{R}^2$

tan

Exponential $\mathbb{R}, \mathbb{R}^2$

Sin, Cos $\mathbb{R}, \mathbb{R}^2$

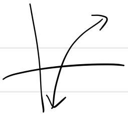
Tan $\neq \frac{\pi}{2}, \frac{3\pi}{2},$

Ran: polynomial

$\underbrace{x^2+y^2}_{\geq 0}+1 \quad \{z \in \mathbb{R} \mid z \geq 1\}$

$x^2 + 4x^2y^6 \geq 0 \quad \{z \in \mathbb{R} \mid z \geq 0\}$

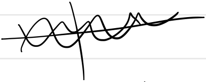
Logarithms



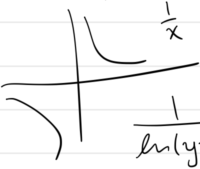
$(-\infty, \infty)$ $\mathbb{R}$

$(0, \infty)$

$[-1, 1]$



$\frac{1}{x}$ $(-\infty, 0) \cup (0, \infty)$



Dom:

$$\frac{1}{\ln(y-x^2)}$$

$\ln(y-x^2)$

$y-x^2 > 0$

$\ln(y-x^2) \neq 0$

$y-x^2 \neq 1$

$\ln x = 0 \Rightarrow x = 1$

$\frac{1}{\ln(y-x^2)}$

Dom: $\{(x,y) \in \mathbb{R}^2 \mid y - x^2 \geq 0,\ y - x^2 \neq 1\}$

Ran: $(-\infty, 0) \cup (0, \infty)$

$\ln(y-x^2)$

$y - x^2 \geq 0$

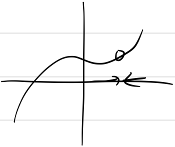
$\ln(y-x^2) \neq 0$

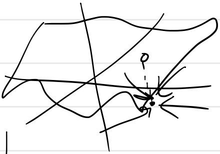
$y - x^2 \neq 1$

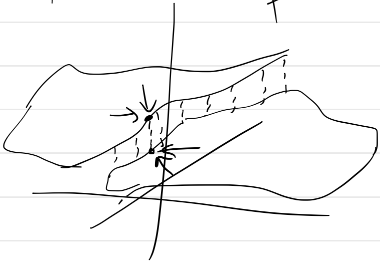
$\ln(y-x^2) \neq 0$

$\frac{1}{\ln(y-x^2)} \neq 0$

10.2 Limits and Continuity







$$\lim_{(x,y)\to(0,0)} \frac{x^2-2y^2}{x^2+y^2} = \lim_{(x,y)\to(0,0)} \frac{0-0}{0-0} = \frac{0}{0} = \boxed{DNE}$$

positive x-axis $\Rightarrow y=0$ ①

positive y-axis $\Rightarrow x=0$ ②

① $\lim_{x\to0^+} \frac{x^2-2(0)^2}{x^2+(0)^2} = \lim_{x\to0^+} \frac{x^2}{x^2} = \lim_{x\to0^+} 1 = 1$

② $\lim_{y\to0^+} \frac{0^2-2y^2}{0^2+y^2} = \lim_{y\to0^+} \frac{-2y^2}{y^2} = \lim_{y\to0^+} -2 = -2$

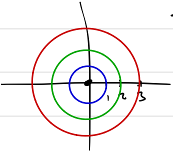
$\neq$

③ along the line $y=x$

along the line $y=x^2$

$C = 0, 1, 2, 3$

$f(x,y) = \sqrt{x^2+y^2}$ Ran: $\{z \in \mathbb{R} \mid z \geq 0\}$



- $0 = \sqrt{x^2+y^2}$
- $1 = \sqrt{x^2+y^2} \Rightarrow 1 = x^2+y^2$
- $2 = \sqrt{x^2+y^2} \Rightarrow 4 = x^2+y^2$
- $3 = \sqrt{x^2+y^2} \Rightarrow 9 = x^2+y^2$